01 QUESTION ONE · MULTIMEDIA BASICS

Define Multimedia. What are the components of Multimedia? What are the different applications of Multimedia?

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DEFINITION OF MULTIMEDIA

The word **multimedia** is made of two parts — *multi*, meaning many, and *media*, meaning the ways in which information is presented. Put together, multimedia means the use of **more than one medium together** in a single presentation. In computing, multimedia is more precisely defined as the **integration of text, graphics, images, sound, video and animation** inside one digital system, where the user is usually able to **interact** with the content rather than only watch it. That last word, interaction, is what separates true multimedia from a plain film or a printed book.

A typical multimedia application lets the user click, choose, type or drag, and the content responds. When a student clicks a diagram in an educational CD, a voice explains that part, an animation shows how it works and a short quiz appears at the end — all inside one program. So many media together make the message richer, because each medium carries what it does best: a graph explains numbers, a sound explains a tone, a video explains a process, text explains the exact wording.

Linear and Non-linear Multimedia

Multimedia is of two kinds. **Linear multimedia** runs from start to finish without control from the user, such as a presentation played in a hall. **Non-linear multimedia** (interactive multimedia) gives the user control, so the user decides what to see and when — a website, a game or an e-learning package. Modern multimedia is almost always non-linear, which is what makes it such a powerful teaching and business tool.

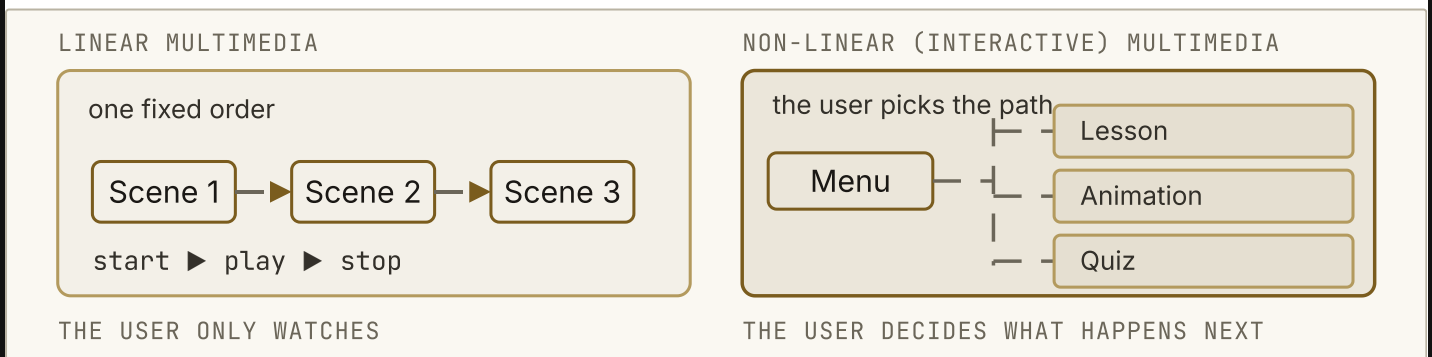


FIG. 1 — TWO KINDS OF MULTIMEDIA. LINEAR CONTENT MUST BE WATCHED IN THE ORDER IN WHICH IT WAS MADE; INTERACTIVE CONTENT HANDS THE CONTROL TO THE USER, WHO JUMPS FROM THE MENU TO WHATEVER INTERESTS THEM.

COMPONENTS OF MULTIMEDIA

A multimedia system is built from six standard components. Each one is a full field of study in its own right, but in a multimedia application they are combined so that the whole becomes more useful than the parts.

- **Text** — the written word, such as headings, paragraphs and captions. It is the oldest and still the most exact medium, because nothing else states a fact as clearly. Text is stored as ASCII, Unicode or in HTML.
- **Graphics and Images** — drawings, photographs, icons and charts. Graphics are of two types — *bitmap* (made of pixels, e.g. a photo) and *vector* (made of lines and shapes, e.g. a logo). They are stored as JPEG, PNG, GIF or SVG.
- **Audio** — any sound, such as speech, music or an alert tone. Audio adds emotion and clarity, and is stored as MP3, WAV or AAC.
- **Video** — moving pictures with sound, used to show real events and processes. Video is heavier than the other media and is stored as MP4, AVI or MKV.
- **Animation** — a sequence of drawings shown quickly so that the picture appears to move. Animation explains processes that a still picture cannot, and is stored as GIF, MP4 or inside CSS.
- **Interactivity** — the element that reacts to the user through clicks, touches, typing or voice. It is the component that binds all the others into one experience.

COMPONENT	EXAMPLE	COMMON FORMATS
Text	Headings, captions	TXT, HTML, PDF
Image	Photo, icon, chart	JPEG, PNG, GIF, SVG
Audio	Narration, music	MP3, WAV, AAC
Video	Clip, film	MP4, AVI, MKV
Animation	Moving logo	GIF, MP4, CSS
Interactivity	Buttons, quizzes	HTML, JS, Flash

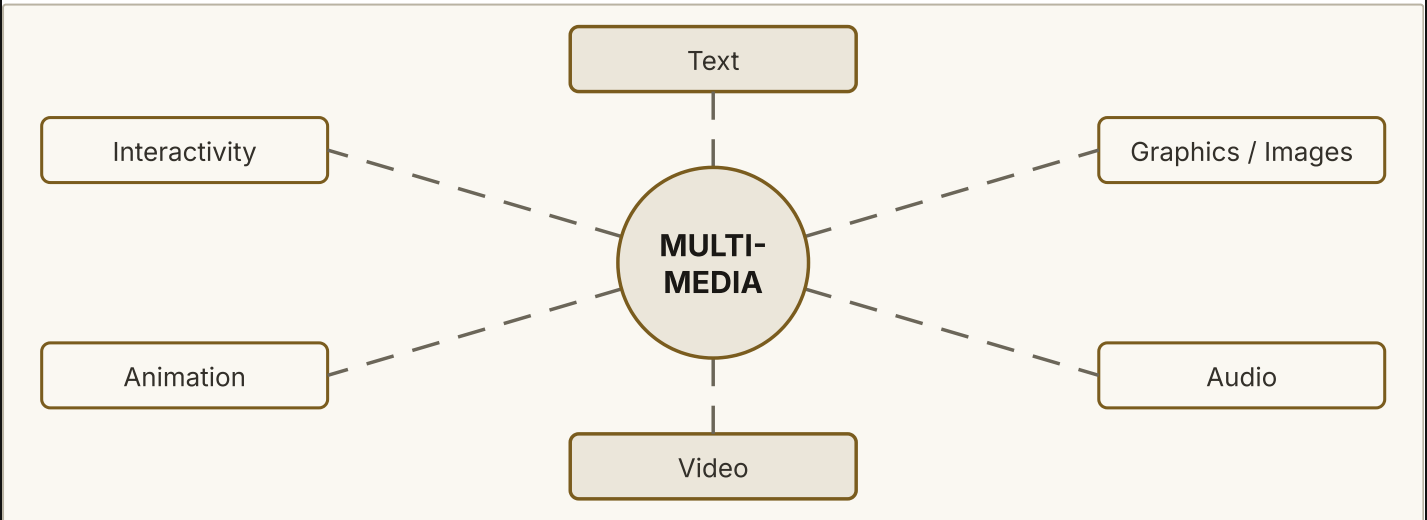


FIG. 2 — THE SIX COMPONENTS OF MULTIMEDIA. THE CENTRE IS NOT A SEVENTH MEDIUM; IT IS THE COMBINATION THAT TURNS SEPARATE MEDIA INTO ONE INTERACTIVE EXPERIENCE.

APPLICATIONS OF MULTIMEDIA

Multimedia is used wherever information has to be presented clearly, attractively and interactively, and it has therefore entered almost every field of modern life. The main applications are described below.

(a) Education and E-Learning

Multimedia has changed the classroom completely. A topic such as the water cycle can be taught with an animation, a diagram, narration and a short quiz, and the same lesson can be played again and again by each student. Smart classes, online courses, language-learning apps and virtual laboratories all depend on it. Because both the eyes and the ears are engaged at once, the student remembers far more than through reading alone.

(b) Entertainment

Films, video games, streaming music and streaming video are the most visible use of multimedia. Computer-generated effects, background music, animation and interactivity are combined in games and films to create experiences that would be impossible in any single medium.

(c) Business, Marketing and Advertising

Companies use multimedia in advertisements, product demos, corporate presentations, video conferences and e-commerce websites. A product video with sound and animation sells far better than a printed leaflet, and a presentation with charts reaches the audience faster than a long report.

(d) Medicine and Healthcare

Doctors use multimedia for training (surgery simulations, anatomy animations), for diagnosis (ultrasound, CT and MRI images) and for telemedicine, where a specialist in one city guides a doctor in another through a live video link. Multimedia has made medical knowledge both easier to teach and easier to share.

(e) Journalism and Broadcasting

News is now delivered as text, photographs, audio clips and live video on one screen. Weather forecasts, cricket scorecards and election results are all illustrated with graphics and animation within a single web page.

(f) Engineering, Design and Architecture

Engineers use CAD and 3-D models, architects show walk-throughs of buildings, and designers preview products before they are made. Multimedia allows a design to be seen and tested before it is built.

(g) Government and Public Services

E-governance, digital museums, tourism kiosks and public-information screens use multimedia to explain schemes and services to citizens in a simple, visual way.

(h) Virtual Reality, Gaming and Emerging Technology

VR headsets, AR apps and modern games combine every component of multimedia at once to place the user inside a fully interactive world.

HTML EXAMPLE

A SINGLE WEB PAGE THAT BRINGS TEXT, IMAGE, AUDIO AND VIDEO TOGETHER — THE WHOLE IDEA OF MULTIMEDIA IN ONE FILE.

MULTIMEDIA.HTML

```html
<!DOCTYPE html>
<html>
<head><title>My Multimedia Page</title></head>
<body>
  <h1>Multimedia on a Single Page</h1>
  <p>This page mixes text, image, sound and video.</p>

  <!-- 1. image -->
  <img src="college.jpg" alt="College photo" width="320">

  <!-- 2. audio -->
  <audio controls>
    <source src="song.mp3" type="audio/mpeg">
  </audio>

  <!-- 3. video -->
  <video width="320" controls>
    <source src="clip.mp4" type="video/mp4">
  </video>
</body>
</html>
```

CONCLUSION

Multimedia is the integration of text, image, audio, video, animation and interactivity into one digital experience. Its six components give it richness, and its applications cover education, entertainment, business, medicine, journalism, engineering and government. Wherever information must be shown clearly and enjoyed, multimedia is the tool of choice.

02 QUESTION TWO · IMAGE FILE FORMATS

Discuss the image file formats and explain the differences between them.

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INTRODUCTION TO IMAGE FILE FORMATS

An **image file format** is the standard way a picture is stored on disk. The format decides how colours are written down, whether the file is compressed, whether it supports transparency, and which programs can open it. This matters because the same picture may be 40 KB in one format and 4 MB in another, and that difference decides how fast a web page loads. Images fall into two families. A **raster** (bitmap) image is a grid of coloured dots called pixels; enlarged too far, the pixels become visible and the picture looks blocky. A **vector** image is described by mathematical lines and shapes, so it enlarges to any size without loss of quality. Raster formats suit photographs, vector formats suit logos, icons and diagrams.

There is also an important difference in compression. **Lossless** compression makes a file smaller without losing a single detail, so the picture is restored exactly. **Lossy** compression makes the file much smaller by throwing away detail the human eye is unlikely to notice, and that detail can never be recovered.

1. JPEG (JOINT PHOTOGRAPHIC EXPERTS GROUP)

JPEG is the most widely used format for **photographs**, because it uses lossy compression and can reduce a photo to a small fraction of its original size. It supports up to 16.7 million colours, which is why photographs look natural in it. Its weaknesses are that it does not support transparency, and that every re-save loses a little more quality.

2. PNG (PORTABLE NETWORK GRAPHICS)

PNG was created as a free replacement for GIF. It uses **lossless** compression, so no quality is lost, and it supports **transparency**. It is ideal for icons, logos, screenshots and images needing a transparent background; the only price is a larger file than JPEG for photographs.

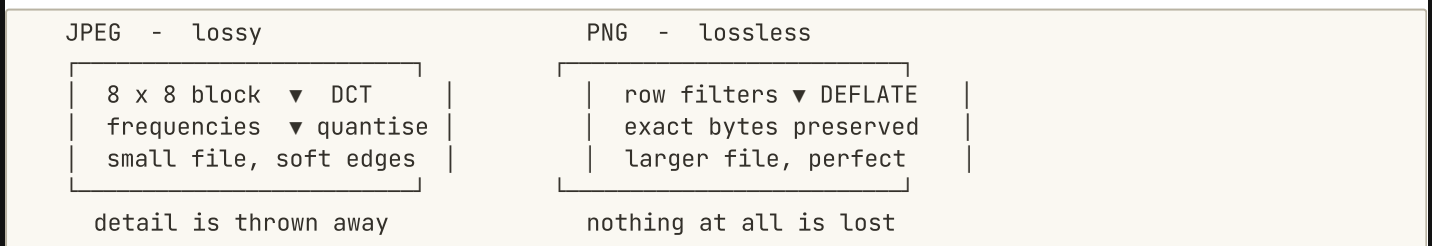


FIG. 1 — THE TWO COMPRESSION IDEAS SIDE BY SIDE, IN TEXT FORM. JPEG TURNS EACH SMALL BLOCK OF PIXELS INTO FREQUENCIES AND THEN DELETES THE WEAK ONES, WHICH IS WHY IT IS SMALL; PNG FILTERS EACH ROW OF PIXELS AND THEN APPLIES A LOSSLESS DICTIONARY, WHICH IS WHY IT IS EXACT.

3. GIF (GRAPHICS INTERCHANGE FORMAT)

GIF is an old but still useful format. It uses lossless compression but is limited to **256 colours**, so it is poor for photographs yet perfect for logos, cartoons and simple graphics. Its best-known feature is that it can store **several frames in one file**, making a GIF a small animation. It also supports **one** transparent colour.

4. BMP (BITMAP)

BMP is a very simple format in which each pixel is stored almost exactly as it appears. It is easy for programs to read, but uses little or no compression, so BMP files are extremely large. It is used mainly inside Windows, not for the web.

5. TIFF (TAGGED IMAGE FILE FORMAT)

TIFF is a flexible, lossless format used by professionals — printing houses, photographers and scanners. It preserves full quality and stores extra information (tags) such as camera settings, but at the cost of a very large file.

6. WEBP

WebP is a modern format developed by Google for the web. It supports both **lossy and lossless** compression plus transparency and animation, and produces much smaller files than JPEG or PNG. Only very old software may fail to open it.

7. SVG (SCALABLE VECTOR GRAPHICS)

SVG is a **vector** format described in XML. Because it is made of shapes and not pixels, it enlarges to any size with no loss of quality, and its files are very small. It is perfect for logos, icons and charts, but cannot store a real photograph.

8. RAW

A RAW file stores exactly what the camera sensor recorded, with almost no processing, so a photographer can adjust the picture fully later. These files are the largest of all and each camera brand has its own version, so RAW is used only in professional work.

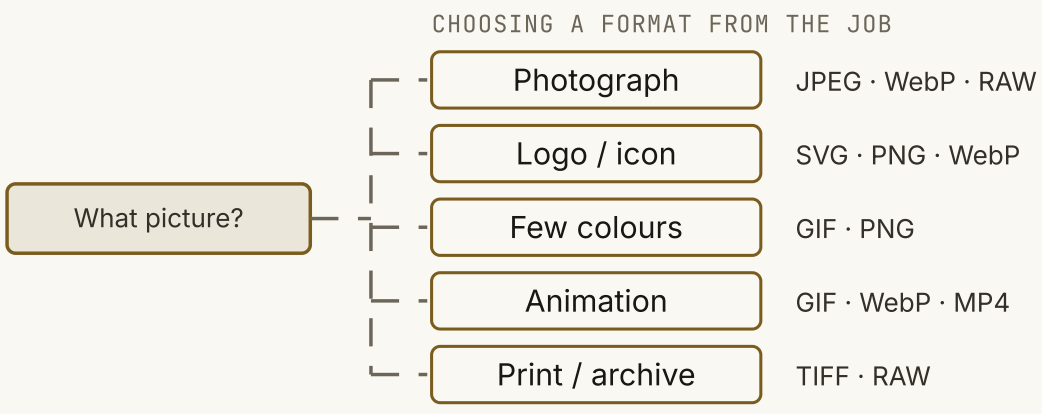


FIG. 2 — A QUICK CHOOSER. THE RIGHT FORMAT IS DECIDED BY THE JOB, NOT BY FASHION: PHOTOGRAPHS WANT JPEG OR WEBP, LOGOS WANT SVG, FLAT CARTOONS WANT GIF, AND PRINT WANTS TIFF OR RAW.

DIFFERENCES BETWEEN THE IMAGE FORMATS

FORMAT	TYPE	COMPRESSION	TRANSPARENCY	BEST USED FOR
JPEG	Raster	Lossy	No	Photographs, web images
PNG	Raster	Lossless	Yes	Icons, logos, screenshots
GIF	Raster	Lossless	1 colour	Simple graphics, small animation
BMP	Raster	None / weak	No	Windows, editing
TIFF	Raster	Lossless	Yes	Printing, scanning, archival
WebP	Raster	Lossy / lossless	Yes	Fast modern web images
SVG	Vector	Lossless (text)	Yes	Logos, icons, charts
RAW	Raster	None	No	Professional photography

The differences can be placed in four groups. **By compression:** JPEG and WebP (lossy mode) throw away detail to become small; PNG, GIF, TIFF and SVG keep every detail. **By colour range:** JPEG and PNG handle millions of colours, while GIF handles only 256. **By transparency:** PNG, WebP, SVG and TIFF support it, JPEG and BMP do not, and GIF supports only one transparent colour. **By scalability:** only the vector format SVG can be enlarged without any loss.

In practice, a web designer chooses JPEG for a photograph, PNG for a logo with a transparent background, GIF for a small animation, SVG for an icon that must stay sharp on every screen, and WebP when both small size and quality are required. A printing shop, on the other hand, chooses TIFF to protect quality.

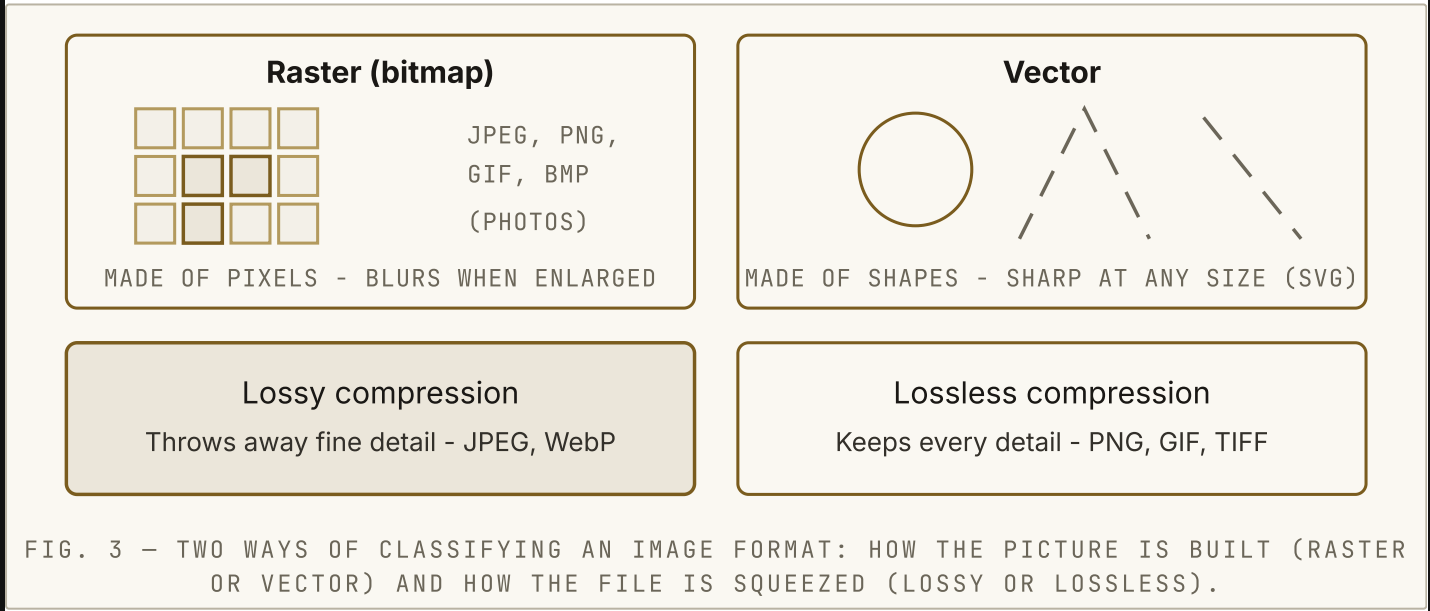


FIG. 3 — TWO WAYS OF CLASSIFYING AN IMAGE FORMAT: HOW THE PICTURE IS BUILT (RASTER OR VECTOR) AND HOW THE FILE IS SQUEEZED (LOSSY OR LOSSLESS).

PYTHON PROGRAM

THE PROGRAM BELOW OPENS ONE IMAGE AND SAVES IT IN SEVERAL FORMATS, SHOWING HOW THE SAME PICTURE CAN BE STORED IN DIFFERENT WAYS. IT USES THE PILLOW (PIL) LIBRARY. COPY THIS ON PAGE 4.

IMAGE_FORMATS.PY

```python
# image_formats.py - open one image and save it in many formats
from PIL import Image

# 1. open a JPEG photograph
img = Image.open("photo.jpg")
print("Opened :", img.format)      # JPEG
print("Size   :", img.size)        # (width, height) in pixels
print("Mode   :", img.mode)        # RGB

# 2. save as PNG (lossless, supports transparency)
img.save("photo.png")
print("Saved  : photo.png  (lossless)")

# 3. make a thumbnail and save as GIF (256 colours)
img.thumbnail((200, 200))
img.convert("P").save("photo.gif")
print("Saved  : photo.gif  (256 colours)")

# 4. create a new image and save as BMP (uncompressed)
new = Image.new("RGB", (100, 100), "navy")
new.save("block.bmp")
print("Saved  : block.bmp  (uncompressed)")
```

Output of the Program

OUTPUT

```
Opened : JPEG
Size   : (800, 600)
Mode   : RGB
Saved  : photo.png  (lossless)
Saved  : photo.gif  (256 colours)
Saved  : block.bmp  (uncompressed)
```

CONCLUSION

An image file format decides how a picture is stored, compressed and opened. JPEG is best for photographs, PNG for transparent graphics, GIF for small animations, TIFF for printing, WebP for the modern web and SVG for sharp scalable artwork. The right format is chosen according to the purpose, the quality needed and the size that can be allowed.

What is an animation?

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CSS ANIMATION ON PAGE 4

DEFINITION OF ANIMATION

Animation is the technique of creating an illusion of movement by showing a series of still pictures one after another very quickly. Each picture is slightly different, and when they are displayed fast enough the eye joins them into one smooth motion — so animation is not a moving picture, it is many still pictures played at speed. The reason this works is a property of human vision called the **persistence of vision**: the brain keeps an image for a very short fraction of a second after it has gone, so if the next image arrives before that fraction passes, the two merge and no gap is noticed. In practice about **24 frames per second** or more looks perfectly smooth — the same principle on which cinema and television have always worked.

Animation should not be confused with video. Video records real, continuous events with a camera; animation **creates** the frames artificially, frame by frame. That gives it one great advantage: anything that can be imagined can be brought to life, even if it can never really happen. In a multimedia title animation is used in two ways — as decoration, such as a moving logo, and as explanation, such as a process broken into steps. It can show the inside of an atom, the centre of the earth, a plant growing over a hundred years, or a machine that does not yet exist — anything hidden, too slow, too fast or too dangerous to film.

On a computer an animation is stored as a long sequence of frames, each one very like an ordinary image. Because a single second may hold twenty-four or more of them, animation files are naturally heavy, and the animator's work is not finished until the file is small enough to run smoothly. Modern tools and formats — GIF, MP4 and CSS animations — exist mainly to solve this one problem: smooth movement in a file small enough to send, stream or open quickly.

TYPES OF ANIMATION

- **Traditional (2-D) animation** — every frame is drawn by hand on paper or a tablet. This is the oldest type and gives a warm, artistic look, but it needs an enormous amount of work. Old Disney films are the classic example.
- **2-D computer animation** — the same idea done on a computer, where objects are drawn once and then moved. It is much faster than hand drawing and is used in most modern cartoons and explainer videos.
- **3-D computer animation** — objects are built as three-dimensional models and lit, textured and moved by a computer. It is used in modern films, games and product walk-throughs and looks the most realistic.
- **Stop-motion animation** — real objects (clay figures, dolls or paper) are photographed one position at a time, and the position is changed slightly between shots. When the photographs are played together the object appears to move by itself.
- **Motion graphics** — text, logos and simple shapes are moved to explain an idea. It is used in advertisements, news graphics and title sequences.
- **Procedural / web animation** — the movement is produced by a program or a style rule rather than drawn by hand. CSS and JavaScript animations on web pages, and particle effects in games, belong to this type.

How Animation is Created

Whatever the type, creating an animation passes through the same stages. First a **storyboard** is prepared, drawing the whole sequence as rough sketches so the story is clear before the expensive work begins. Then the **keyframes** are drawn — the few important positions of each movement — and the frames in between are filled in by **tweening**, which the computer now performs automatically, whereas in the past an artist had to draw every single one. After this the frames are **rendered** with colour, light and texture, the sound is added, and the sequence is **exported** as a GIF, a video file or a CSS animation. The quality of the result depends mostly on the skill of the keyframes and the smoothness of the tweening, because the eye judges movement by how evenly the frames flow, not by how many there are.

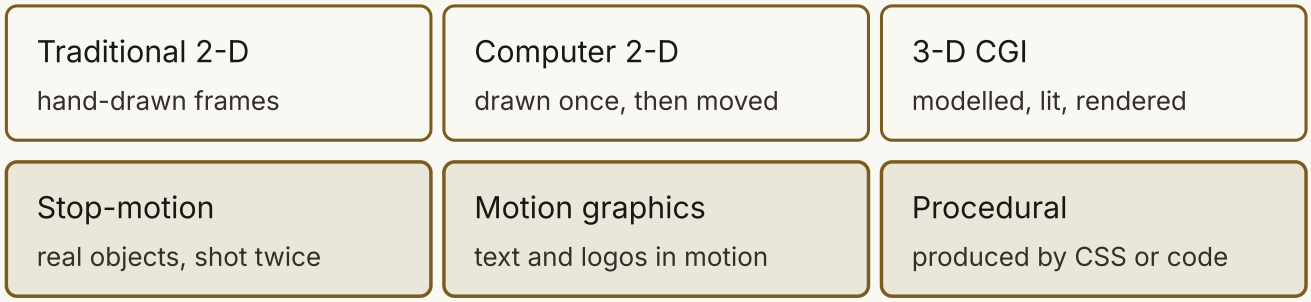


FIG. 1 — THE SIX TYPES OF ANIMATION. THE TOP ROW IS BUILT BY AN ARTIST, THE BOTTOM ROW IS CAPTURED FROM THE REAL WORLD OR PRODUCED BY CODE; EVERY TYPE ENDS AS THE SAME THING — A NUMBERED ROW OF FRAMES.

KEY TERMS RELATED TO ANIMATION

- **Frame** — a single still picture out of the many that make up an animation.
- **Frame rate (fps)** — the number of frames shown in one second. The higher the rate, the smoother the motion.
- **Keyframe** — an important frame that the animator sets by hand; the computer then fills in the frames between two keyframes.
- **Tweening** — the automatic filling of frames between two keyframes, so the animator does not have to draw every single frame.
- **Onion skinning** — showing the previous frame faintly underneath the current one, so the artist can match the movement.
- **Storyboard** — a rough set of sketches that plans the whole animation before the real work begins.

USES AND APPLICATIONS OF ANIMATION

Animation is used in **entertainment** (cartoons, films, games), in **education** (showing a process step by step), in **advertising** (an animated banner catches the eye far better than a still one), in **science and medicine** (a beating heart, a chemical reaction), in **engineering** (a cut-away moving shot of an engine), in **user interfaces** (a loading spinner, a button that slides) and in **websites** (icons that move when touched). It can also present something dangerous — a blast, an accident, a surgical step — safely, again and again.

Two practical matters are worth remembering. The first is the **frame rate**: below about 12 fps the movement looks jerky, 24 fps looks natural, and 50–60 fps looks very smooth but makes the file much heavier. The second is **file size**: since an animation is a long line of pictures, it becomes very large unless the frames are compressed — which is why animated GIFs are limited to 256 colours and why video files use modern codecs. A good animator balances three things at once: the smoothness of the movement, the sharpness of the picture and the size of the file.

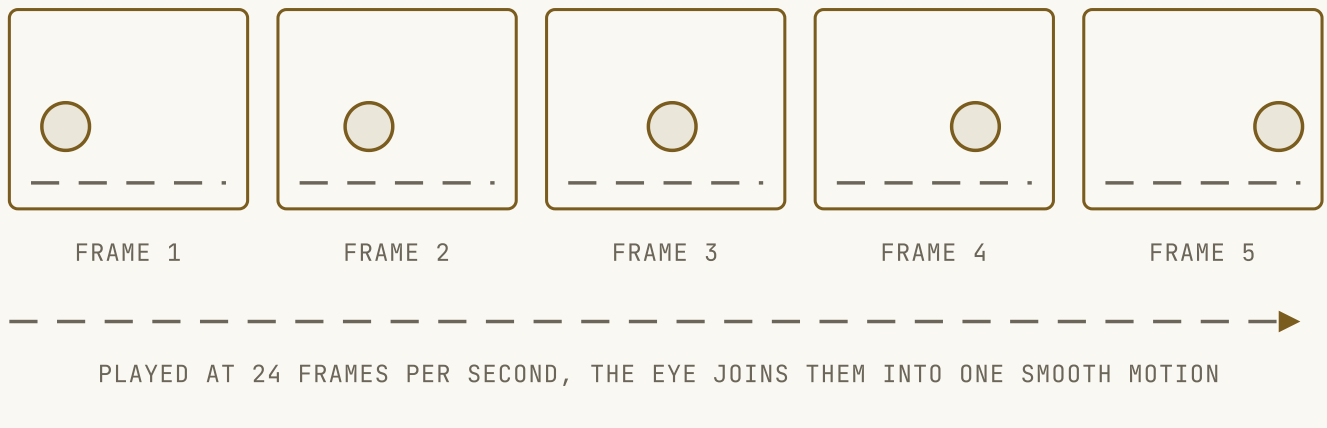


FIG. 2 — PERSISTENCE OF VISION. EVERY FRAME IS A STILL PICTURE; THE BRAIN BRIDGES THE GAPS WHEN THE FRAMES ARRIVE FAST ENOUGH (ABOUT 24 PER SECOND OR MORE).

CSS ANIMATION EXAMPLE

THE PAGE BELOW CREATES A MOVING SQUARE USING ONLY CSS — `@KEYFRAMES` DEFINES THE MOVEMENT AND THE `ANIMATION` PROPERTY PLAYS IT. COPY THIS ON PAGE 4.

ANIMATION.HTML

```
<!DOCTYPE html>
<html>
<head>
<style>
  .box {
    width: 80px; height: 80px;
    background: #7a5c1e;
    position: relative;
    animation: slide 3s infinite alternate;
  }
  /* the movement is defined here */
  @keyframes slide {
    from { left: 0;     background: #7a5c1e; }
    to   { left: 300px; background: #2f5d3a; }
  }
</style>
</head>
<body>
  <h1>CSS Animation Demo</h1>
  <div class="box"></div>
  <p>The square slides right and changes colour.</p>
</body>
</html>
```

How the Example Works

The rule `@keyframes slide` defines two keyframes — *from* (the starting position) and *to* (the ending position). The property `animation: slide 3s infinite alternate` then tells the browser to run those keyframes for three seconds, to repeat them forever, and to move forwards and backwards on each round. The browser fills every frame in between automatically, which is exactly the idea of tweening.

CONCLUSION

Animation is the art of producing movement from a series of still frames, made possible by the persistence of vision at about 24 frames per second. It appears in many types — traditional, 2-D, 3-D, stop-motion, motion graphics and procedural — and it is used wherever a process, an idea or an emotion has to be shown rather than merely described. In multimedia it is used both to decorate and to explain, and it can show even the impossible perfectly.